ENTRALENS

Intelligent Identity Security Copilot for Microsoft Entra ID

Project Submission Report

1. Introduction

Modern organizations use Microsoft Entra ID to manage users, permissions, authentication, licenses, and security policies. As the number of users increases, administrators face challenges in identifying inactive accounts, monitoring MFA compliance, managing privileged accounts, and investigating security risks.

Traditional systems require administrators to manually write SQL queries and analyze database records, which is time-consuming and requires technical expertise.

To address this challenge, the EntraLens project introduces an AI-powered Identity Security Copilot that enables administrators to interact with identity security data using natural language in both English and Bangla.

The main focus of this project is AI Intent Detection and Automatic SQL Query Generation.

2. Problem Statement

Identity administrators often need answers to questions such as:

- Which users have not logged in for 90 days?
- Which accounts do not have MFA enabled?
- Which department has the highest risk score?
- Which privileged accounts require investigation?

Normally, answering these questions requires manual SQL queries and database expertise.

The objective of EntraLens is to eliminate this complexity by allowing administrators to ask questions naturally while the AI automatically understands the intent, generates SQL queries, executes them against the database, and returns results.

3. Project Objectives

The main objectives of the project are:

1. Detect user intent from natural language questions.
2. Generate secure SQL queries automatically.
3. Execute queries against PostgreSQL.
4. Display results through an interactive dashboard.
5. Support both text and voice interaction.
6. Support Bangla and English queries.
7. Provide identity security insights for administrators.

4. System Overview

EntraLens consists of four major components:

Frontend Layer

Provides the user interface where administrators can:

- Ask questions
- Use voice commands
- View generated SQL
- View query results
- Manage users

AI Processing Layer

Responsible for:

- Intent Detection
- Natural Language Understanding
- SQL Query Generation

Backend Layer

Implemented using FastAPI.

Responsibilities:

- Process requests
- Communicate with AI model
- Execute SQL queries
- Return results

Database Layer

PostgreSQL stores:

- Users
- Roles
- Sign-in Logs
- Audit Events
- Security Data

5. AI Intent Detection

AI Intent Detection is the core feature of the project.

The purpose of intent detection is to identify what information the administrator wants from the system.

Example 1

User Question:

Show inactive users

Detected Intent:

Find inactive accounts

Example 2

User Question:

MFA enable না থাকা user দেখাও

Detected Intent:

Find users without MFA

Example 3

User Question:

Finance department-এর user দেখাও

Detected Intent:

Retrieve users from Finance department

The AI model analyzes the meaning of the user's request instead of relying on predefined buttons.

This allows the system to understand different wording while producing the same result.

6. AI SQL Query Generation

After detecting the intent, the AI automatically generates a PostgreSQL query.

Example

User Question:

Show inactive users

Generated SQL:

```
SELECT * FROM users
WHERE last_login_days >= 90;
```

Example

User Question:

Finance department-এর user দেখাও

Generated SQL:

```
SELECT * FROM users
WHERE LOWER(department)=LOWER('Finance');
```

Example

User Question:

Show high risk users

Generated SQL:

```
SELECT * FROM users
WHERE risk_score >= 70
ORDER BY risk_score DESC;
```

This eliminates the need for administrators to write SQL manually.

7. Bilingual AI Support

A unique feature of EntraLens is bilingual support.

The system accepts:

English

Show Finance users

Bangla

ফাইন্যান্স ডিপার্টমেন্টের ইউজার দেখাও

Mixed Language

Finance department-এর user দেখাও

Regardless of language, the AI identifies the same intent and generates the correct SQL query.

8. Voice-to-SQL Architecture

The project also supports voice interaction.

Workflow:

Administrator speaks a question

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Speech Recognition converts speech into text

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AI detects intent

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AI generates SQL

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PostgreSQL executes query

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Results are displayed

This allows administrators to perform investigations without typing.

9. Live Database Query Execution

Generated SQL queries are executed against PostgreSQL.

Workflow:

User Question

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Intent Detection

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SQL Generation

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FastAPI Backend

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PostgreSQL Execution

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Results Returned

Only SELECT queries are allowed to prevent database damage.

Dangerous commands such as:

- DELETE
- DROP
- ALTER
- TRUNCATE
- UPDATE

are blocked.

10. Database Design

The EntraLens platform uses PostgreSQL as its primary database. The database is designed to store identity information, security events, role assignments, and sign-in activities required for AI-powered security analytics and SQL query generation.

Table 1: Users

Column Name	Data Type	Description
id	Integer (PK)	Unique user identifier
tenant_id	VARCHAR(50)	Tenant identifier
display_name	VARCHAR(100)	User full name
email	VARCHAR(150)	User email address
department	VARCHAR(100)	User department
user_type	VARCHAR(50)	User category
mfa_enabled	BOOLEAN	MFA status
license_type	VARCHAR(20)	Assigned license
last_login_days	INTEGER	Days since last login
risk_score	INTEGER	Calculated security risk score
account_status	VARCHAR(20)	Active/Terminated status

Table 2: Role Assignments

Column Name	Data Type	Description
id	Integer (PK)	Unique role record ID
user_email	VARCHAR(150)	User email
role_name	VARCHAR(100)	Assigned role
privileged	BOOLEAN	Privileged role indicator
assigned_date	DATE	Role assignment date

Table 3: Sign-In Logs

Column Name	Data Type	Description
id	Integer (PK)	Log identifier
user_email	VARCHAR(150)	User email
login_time	TIMESTAMP	Login timestamp
ip_address	VARCHAR(50)	Source IP address
country	VARCHAR(100)	Login location
status	VARCHAR(20)	Success or failure

Table 4: Audit Events

Column Name	Data Type	Description
id	Integer (PK)	Event identifier
event_type	VARCHAR(100)	Event category
user_email	VARCHAR(150)	User associated with event
description	TEXT	Event details
created_at	TIMESTAMP	Event creation time

Entity Relationship Overview

The database follows a security-centric design:

- The **Users** table stores identity information.
- The **Role Assignments** table stores user roles and privileged access information.
- The **Sign-In Logs** table stores authentication activities.
- The **Audit Events** table stores administrative and security-related actions.

These tables collectively provide the data source for:

- AI Intent Detection
- SQL Query Generation
- Identity Security Analytics
- Risk Assessment
- User Management
- Security Investigation Workflows

11. Security Features

The project includes Role-Based Access Control (RBAC).

Global Administrator

Full access to all modules.

IAM Engineer

Access to:

- User Management
- Identity Governance

Billing Administrator

Access to:

- License Optimization
- Inactive Users
- Orphan Accounts

Unauthorized Users

Access denied.

12. System Architecture

Natural Language Input

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AI Intent Detection

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SQL Query Generation

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FastAPI Processing

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PostgreSQL Query Execution

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Dashboard Visualization

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Security Insights

13. Technologies Used

Frontend:

- HTML
- CSS
- JavaScript
- Chart.js

Backend:

- Python 3.13
- FastAPI
- SQLAlchemy

Database:

- PostgreSQL

AI:

- Ollama
- Gemma / Llama Models

Voice Processing:

- Web Speech API

Version Control:

- Git
- GitHub

14. Results

The system successfully demonstrates:

- AI Intent Detection
- Automatic SQL Query Generation
- PostgreSQL Query Execution
- Bangla and English Query Support
- Voice-to-SQL Processing
- Identity Security Analytics

The project reduces the complexity of database investigation and allows administrators to retrieve security information through natural language.

15. Conclusion

EntraLens demonstrates how Artificial Intelligence can simplify identity security management. Instead of requiring technical SQL knowledge, administrators can interact with the system using natural language and voice commands.

The key innovation of the project is the combination of:

- AI Intent Detection
- SQL Query Generation
- Live PostgreSQL Query Execution

This approach improves productivity, reduces investigation time, and provides a modern AI-powered identity security experience.

Future versions of the project can integrate Microsoft Graph API, Azure Entra ID, Vector Databases, and Advanced RAG architectures to provide enterprise-grade identity security intelligence.